Rupali Tasnim Samad

Dhaka, Bangladesh

RESEARCH INTEREST

Human Computer Interaction, Responsible AI, Applied NLP

EDUCATION

University of Dhaka - Bachelors of Science in Software Engineering(BSSE), Graduated in **Feb 2025**

Accomplishment: Awarded Govt Scholarship for academic excellence in undergraduate program.

PUBLICATIONS

Seeing Like Datafied Healthcare: Rupali Tasnim Samad, Shah Md Farid Uddin, Syed Ishtiaque Ahmed, Sharifa Sultana, **. Manuscript under review CHI 2027 (A*)**

Studying about healthcare data in large public hospitals and how these data can be used to leverage the powers of AI in Bangladesh healthcare system.

Bangladesh's AI Readiness: Perspectives from the Academia, Industry, and Government: Sharifa Sultana, **Rupali Tasnim Samad**, Mehzabin Haque, Zinnat Sultana, Zulkarin Jahangir, B M Mainul Hossain, Rashed Mujib Noman, Syed Ishtiaque Ahmed**. Paper accepted into CHI 2026 Poster(A*)**

LLMs are increasingly integrated into software systems, requiring outputs that meet both syntactic and semantic constraints. Drawing from AI safety literature and formal methods, we identify trends, gaps, and propose an initial constraint calculus to safeguard LLM outputs in software.

Bangladesh AI Readiness: Gaps in Curriculum, Infrastructure, and Governance. Sharifa Sultana, **Rupali Tasnim Samad**, Mehzabin Haque, Zinnat Sultana, Zulkarin Jahangir, B M Mainul Hossain, Rashed Mujib Noman, Syed Ishtiaque Ahmed**.** Paper is accepted to **COMPASS 2026**.

Dynamic Test Case Selection (TCS) with LLM. A. Paul, **R. Tasnim**. Extended Abstract, ***Accepted*** in Samsung Best Paper Award May, 2024

I conducted a survey in Samsung R&D Institute, Bangladesh and found TCS needs are dynamic across projects. No existing tools can handle such dynamic TCS needs. So we used LLM for TCS - proposing Knowledge Graph structures and custom embeddings on project vocabs.

PATENT

Filed for patent in Samsung HQ:

1. **An Optimization of dLoRA Architecture** - filed for patent, Samsung R&D Institute, Bangladesh Jun 2026
2. **Automated Face Anonymization for video content** - filed for patent, Samsung R&D Institute, Bangladesh Apr 2026

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RESEARCH EXPERIENCE

University of Toronto, University of Illinois Urbana Champaign | Dr. Syed Ishtiaque Ahmed, Dr. Sharifa Sultana Ela | Responsible AI, AI Readiness

Submitted three papers: one accepted as a CHI poster and another currently under review at ACM COMPASS. Another one is under review in CHI 2027

University of Toronto, Pennsylvania State University | Dr. Syed Ishtiaque Ahmed, Dr. Tanusree Sharma | Privacy in AI, HCI

Working on a paper that investigates privacy risks associated with shared large language models, with a particular focus on contexts in the Global South.

Samsung R&D Institute Bangladesh | Engineer | Biz Solution Group

- Samsung Best Paper Award: Hybrid RAG Personalization was selected as one of the top three ideas from SRBD and nominated to HQ.
- Mobile Developer Conference: Automated Face Anonymization for Live Video Content was the only idea selected from SRBD for submission to the Samsung Mobile Developer Conference of 2025.

- AIPoC Contest– A method to reduce false positives in stress measurement using wearable devices was awarded "People's Champion" in the AIPoC contest, receiving the highest votes from 293 engineers. Owing to its novelty, management recommended pursuing a DOI for patent filing, which is currently in progress.

Monash University | HumaniSE Lab | **Software Engineering** Dec 2022 - Mar 2023
Analyzed their requirements for a software product that translates android app to ensure localization that respects the human value of universalism. We devised a software requirements specification and system design as part of IEEE ICSE. We designed Localization of Android Apk accessibility.

SCHOLARSHIPS

- Received Govt. Scholarship for undergraduate result in University of Dhaka 2025
- Received Govt. Scholarship in Higher Secondary Certification Examination, Bangladesh. 2017-2019

AWARDS

- Received **Excellence in Research Award,** 2025 2nd Quarter, Samsung R&D Institute, Bangladesh. Nov, 2025
- AI Proof-of-Concept (PoC) received the highest number of votes in Samsung R&D Institute, Bangladesh's internal AI PoC contest.. Jul, 2025
- **Champion | UXplorer Quest (UI/UX design contest),** *Cefalo presents ITverse 2023*
- **Champion | HackTheVerse (Hackathon Blockchain Challenge),** *Cefalo presents ITverse 2023*
- **First Runner Up | TechTales (Software Project Showcasing),** *Cefalo presents ITverse 2023*
- **Second Runner Up | INTER UNIVERSITY IDEATHON CONTEST,** *NDUB CSE FEST 2023*
- **Second Runner Up | Project Showcasing,** *IEEE BUP BRANCH FEST 2023*

DEVELOPMENT SKILLSETS

Languages:	Python, JS, Java, R, C, C++, SQL
Libraries and Frameworks:	OpenCV, Scikit-Learn, PyTorch, llama.cpp, React, Next.js, Angular, Flutter, Laravel, FastAPI, Node JS
Development Tools:	Git, Linux/UNIX, Remote Server management, Knowledge Graph

PROJECTS

AirBNB recommendation system
Developed an NLP-driven Airbnb recommendation system leveraging user reviews and textual data to generate personalized property suggestions based on sentiment and preference extraction.

AI Maturity Assessment Tool
LLM based tool for evaluating AI maturity of software companies using **FastAPI, Next.js and MongoDB.**

BSL - Bangla Sign Language Detection & Learning Platform
Interactive Bangla sign language learning website with activity recognition feature using **FastAPI, Angular and Sqlite.**

Earth and Beyond Earth
Immersive educational website exploring countries and solar system using **FastAPI, React and Sqlite.**

DU Bus Tracking System
Real-time bus location display to track the current position of DU buses **using Flutter**

Wumpus World
An intelligent-agent simulation in which an autonomous agent navigates an uncertain environment, reasons from perceptual clues, avoids hazards, and searches for a goal.

Mini Chess
Mini chess game developed using the Next.js framework, where you can play against an AI opponent